Day 020 | Unit 02 | Teacher plan

Positive Rational Numbers: Fractions and Mixed Numbers

Connect Fraction Bars and Related Equations

Standards: 6.NS.A.1

Learning target: I can connect Fraction Bars and Related Equations, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole. Division and multiplication are inverse operations. For T ÷ s = n, the related equation is s × n = T. Label whether the unknown is the total, the group size, or the number of groups.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 15

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01 / Learn

A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole. Division and multiplication are inverse operations. For T ÷ s = n, the related equation is s × n = T. Label whether the unknown is the total, the group size, or the number of groups.

Worked example 1

Draw a bar partitioned into 6 equal parts. Shade 5 parts. Circle groups of 4 parts. Use the model to find 5/6 ÷ 2/3.

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Rename both fractions with a common denominator, draw equal parts, and count groups.

3. Each small part is 1/6

4. 5 shaded parts divided into groups of 4 gives 5/4 = 5/4 groups.

5. State the result in the form requested: 5/4.

Answer: 5/4

Worked example 2

The equation 1/5 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

1. Division and multiplication are inverse operations.

2. 1 ÷ 1/5 = 1 × 5 = 5.

3. Check: 1/5 × 5 = 1.

Answer: n = 5; 1 ÷ 1/5 = 5

Day 020 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a bar partitioned into 12 equal parts. Shade 11 parts. Circle groups of 8 parts. Use the model to find 11/12 ÷ 2/3.

02 / Constructed response

The equation 1/4 × n = 1/2 represents equally sized portions. Find n, write the related division equation, and check.

03 / Constructed response

Draw a bar partitioned into 10 equal parts. Shade 10 parts. Circle groups of 6 parts. Use the model to find 1 ÷ 3/5.

04 / Constructed response

The equation 1/8 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

05 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 3 parts. Circle groups of 3 parts. Use the model to find 3/8 ÷ 3/8.

06 / Constructed response

The equation 1/2 × n = 1/3 represents equally sized portions. Find n, write the related division equation, and check.

07 / Constructed response

Draw a bar partitioned into 6 equal parts. Shade 5 parts. Circle groups of 4 parts. Use the model to find 5/6 ÷ 2/3. Explain the partial group.

08 / Model and constructed response

The equation 2/5 × n = 4/5 represents equally sized portions. Find n, write the related division equation, and check.

09 / Multiple choice

Draw a bar partitioned into 8 equal parts. Shade 7 parts. Circle groups of 4 parts. Use the model to find 7/8 ÷ 1/2. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 7/4
B. 11/4
C. 3/4
D. 35/2

10 / Constructed response

The equation 5/3 × n = 2 represents equally sized portions. Find n, write the related division equation, and check.

Day 020 | Unit 02 | Answer key

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1. 11/8

Each small part is 1/12; 11 shaded parts divided into groups of 8 gives 11/8 = 11/8 groups.

6.NS.A.1 | Constructed response

2. n = 2; 1/2 ÷ 1/4 = 2

1/2 ÷ 1/4 = 1/2 × 4 = 2. Check: 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

3. 5/3

Each small part is 1/10; 10 shaded parts divided into groups of 6 gives 10/6 = 5/3 groups.

6.NS.A.1 | Constructed response

4. n = 8; 1 ÷ 1/8 = 8

1 ÷ 1/8 = 1 × 8 = 8. Check: 1/8 × 8 = 1.

6.NS.A.1 | Constructed response

5. 1

Each small part is 1/8; 3 shaded parts divided into groups of 3 gives 3/3 = 1 groups.

6.NS.A.1 | Constructed response

6. n = 2/3; 1/3 ÷ 1/2 = 2/3

1/3 ÷ 1/2 = 1/3 × 2 = 2/3. Check: 1/2 × 2/3 = 1/3.

6.NS.A.1 | Constructed response

7. 5/4

Each small part is 1/6; 5 shaded parts divided into groups of 4 gives 5/4 = 5/4 groups.

6.NS.A.1 | Constructed response

8. n = 2; 4/5 ÷ 2/5 = 2

4/5 ÷ 2/5 = 4/5 × 5/2 = 2. Check: 2/5 × 2 = 4/5.

6.NS.A.1 | Model and constructed response

9. A - 7/4

Each small part is 1/8; 7 shaded parts divided into groups of 4 gives 7/4 = 7/4 groups.

6.NS.A.1 | Multiple choice

10. n = 6/5; 2 ÷ 5/3 = 6/5

2 ÷ 5/3 = 2 × 3/5 = 6/5. Check: 5/3 × 6/5 = 2.

6.NS.A.1 | Constructed response